GROUP 2602 - The Restricted Boltzmann Machine

Francesco Erario, Andrea Lorenzetto, Laura Mazzilli, and Nicola Zuccotti
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In this work, we implement a Restricted Boltzmann Machine (RBM) following the practical guidelines suggested by Geoffrey Hinton [1]. We train and evaluate the model on a subset of the MNIST dataset (digits 0, 1, and 2). Our implementation incorporates several Hinton-inspired choices: specific activation and updating rules for variance reduction, automated visible bias initialization, and advanced thermodynamic metrics like Annealed Importance Sampling (AIS) to monitor the true log-likelihood. In addition to the standard binary framework, we explore structural extensions such as Ising spins and categorical Potts units. Our results show that even a straightforward implementation of these guidelines allows the RBM to successfully reconstruct handwritten digits and map their energy profiles.

INTRODUCTION

Restricted Boltzmann Machines are generative stochastic neural networks capable of learning a probability distribution over an input dataset. In its baseline configuration, the model is structured as a Binary-Binary RBM, where both the visible layer $\mathbf{v} \in \{0,1\}^{n_v}$ and the hidden layer $\mathbf{h} \in \{0,1\}^{n_h}$ represent Bernoulli random variables.

Following Hinton's formulation [1], the joint energy of a network configuration $(\mathbf{v}, \mathbf{h})$ is defined as:

$$E(\mathbf{v}, \mathbf{h}) = -\sum_{i=1}^{n_v} a_i v_i - \sum_{j=1}^{n_h} b_j h_j - \sum_{i=1}^{n_v} \sum_{j=1}^{n_h} v_i h_j W_{ij} \quad (1)$$

where W_{ij} is the weight connecting visible unit i to hidden unit j, while a_i and b_j represent the visible and hidden biases. The native conditional probabilities are governed by the logistic sigmoid function $\sigma(x) = (1+e^{-x})^{-1}$:

$$P(h_j = 1 \mid \mathbf{v}) = \sigma\left(\sum_{i=1}^{n_v} v_i W_{ij} + b_j\right) \quad (2)$$

$$P(v_i = 1 \mid \mathbf{h}) = \sigma\left(\sum_{j=1}^{n_h} h_j W_{ij} + a_i\right) \quad (3)$$

Continuous Input and Gradient Statistics

Although the standard RBM framework assumes binary variables, the MNIST dataset consists of continuous grayscale pixel intensities normalized in the interval $[0, 1]$. To avoid information loss from hard binarization, we feed these continuous values directly into the visible layer during the positive phase, treating each intensity v_i as the expectation $\mathbb{E}[v_i = 1]$. For the negative phase, our framework supports both Contrastive Divergence ($CD - k$) and Persistent Contrastive Divergence (PCD). Following Hinton's specific guidelines for variance reduction (Section 3.4) [1], we enforce stochastic binarization only when sampling the hidden states to drive the Markov chain. Conversely, the final gradient updates are computed using continuous probabilities rather than sampled binary states, drastically suppressing sampling noise during training.

Model Extensions and Generative Testing

To analyse how different physical constraints affect representations, our framework extends the core architecture to alternative dynamics such has Ising Spins (that converts the system to $\{-1, +1\}$ variables using $tanh$ activations) and Potts Units (that implements categorical multi-class hidden units via a softmax sampling distribution). Once training is complete, the network's generative performance is evaluated. We perform iterative Gibbs sampling ($v \to h \to v$) through a dedicated reconstruction phase, allowing the system to settle into stable configurations and synthesize clean digit profiles from latent states.

Project Objectives

This work targets the following key objectives:

- **Model Characterization:** evaluate baseline performance on MNIST digits ('0', '1', '2') using Hinton's analytical bias initialization.

- **Hyperparameter and Architectural Optimization:** compare SGD with scheduling (parabolic mini-batch growth) against RMSprop, while exploring Binary, Spins, and Potts variants.

- **Energy Landscapes:** Map energy barriers during forced transitions between different digit classes to study the mixing properties and stalling behavior of the Gibbs sampler.

- **Optimal model configuration**: we train the model with the optimized parameters and we try to test its generative capabilities

METHODS

Contrastive Divergence (CD-k) and Gradient Updates

The RBM is trained to maximize the log-likelihood of the data by adjusting its parameters $\theta = \{W, \mathbf{a}, \mathbf{b}\}$ via gradient ascent. The exact gradient of the log-likelihood for a single training sample $\mathbf{v}$ is given by:

$$\frac{\partial \ln P(\mathbf{v})}{\partial W_{ij}} = \langle v_i h_j \rangle_{\text{data}} - \langle v_i h_j \rangle_{\text{model}} \tag{4}$$

where "*data*" denotes the positive phase, and "*model*" represents the expectation under the full model distribution (negative phase), which is analytically intractable. To approximate the negative phase, we implement the Contrastive Divergence ($CD-k$) algorithm initialized with the training data $\mathbf{v}^{(0)} = \mathbf{v}_{\text{batch}}$. Following Hinton's variance reduction techniques [1], the process is structured as follows:

- **Positive Phase:** the hidden states are stochastically sampled using the conditional distribution $P(\mathbf{h} \mid \mathbf{v}^{(0)})$.
- **Intermediate Steps:** for all steps $t < k$, full stochastic binarization is strictly enforced on both layers to preserve the thermodynamic properties of the Markov chain.
- **Final Step:** at step $t = k$, binarization is skipped to suppress sampling noise. The gradient components are updated using the exact continuous probabilities, denoted as $\mathbf{v}^{(k)}$ and $\mathbf{h}^{(k)}$:

$$\Delta W = \frac{1}{N}\left((\mathbf{v}^{(0)})^T \mathbf{h}^{(0)} - (\mathbf{v}^{(k)})^T \mathbf{h}^{(k)}\right) \tag{5}$$

Hinton's Analytical Bias Initialization

To accelerate early optimization, we avoid zero-initialization for the visible biases. Instead, we invert the target activation functions at zero weights ($W = 0$), setting $\mathbf{a}$ to match the empirical mean $\mathbf{p} = \frac{1}{N}\sum \mathbf{v}_i$ of the MNIST pixels. Depending on the network configuration, the mapping is defined as:

- **Bernoulli Units:** The inverse of the logistic sigmoid function maps probabilities to the log-odds space:

$$a_i = \text{logit}(p_i) = \ln\left(\frac{p_i}{1-p_i}\right) \tag{6}$$

- **Ising Spins:** The inverse of the hyperbolic tangent maps the empirical expectation $\mathbf{m} \in [-1, +1]$:

$$a_i = \text{arctanh}(m_i) = \frac{1}{2}\ln\left(\frac{1+m_i}{1-m_i}\right) \tag{7}$$

In practice, values are clipped to prevent numerical divergences ($\pm\infty$).

Evaluation Metrics and Partition Function Estimation

Since the reconstruction Mean Squared Error (MSE) can be an unfaithful indicator of generative quality, we implement other two rigorous thermodynamic metrics:

The tracking of the Free Energy $F(\mathbf{v})$ allows us to monitor overfitting. It is computed for a batch as:

$$F(\mathbf{v}) = -\mathbf{v}^T\mathbf{a} - \sum_{j=1}^{n_h} \ln\left(1 + e^{\mathbf{v}^T \mathbf{W}_{:,j} + b_j}\right) \tag{8}$$

We evaluate the *Free Energy Gap* ($F(\mathbf{v}_{\text{data}}) - F(\mathbf{v}_{\text{fantasy}})$). A divergent gap indicates that the model is assigning unproportionally low energy (high probability) to its own generated states compared to real data.

To compute the exact log-likelihood $\ln P(\mathbf{v}) = -F(\mathbf{v}) - \ln Z$, the intractable partition function Z is estimated using Annealed Importance Sampling (AIS). We construct a geometric schedule of M intermediate inverse temperatures $0 = \beta_0 < \cdots < \beta_M = 1$. The target partition function $\ln Z$ is derived by averaging log-weights $\ln w^{(r)}$ accumulated over R independent sampling runs:

$$\ln Z = \ln Z_{\text{base}} + \ln\left(\frac{1}{R}\sum_{r=1}^{R} e^{\ln w^{(r)}}\right) \tag{9}$$

$$\ln w^{(r)} = \sum_{m=0}^{M-1}\left[F_{\beta_m}(\mathbf{v}^{(m)}) - F_{\beta_{m+1}}(\mathbf{v}^{(m)})\right] \tag{10}$$

where Z_{base} is the analytical partition function of a base model with zero weights ($W = 0$), and F_β represents the intermediate Free Energy.

RESULTS

Here we discuss the results of the analysis of our model based on Hinton's directives. Later, we vary one hyperparameter at a time to study its effect.

Training Analysis

The model is trained for 150 epochs using Contrastive Divergence with L=12 hidden units. Figure 1 summarizes the main training metrics. The free energy of real data samples and of fantasy particles both oscillate over

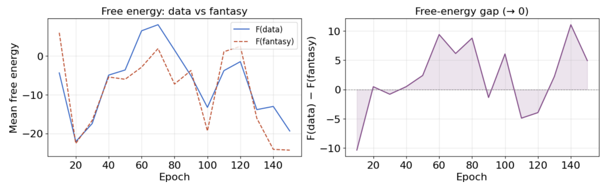

FIG. 1. Training metrics for $L = 12$ hidden units over 150 epochs.

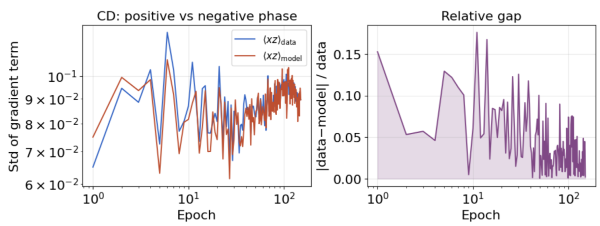

FIG. 2. Contrastive Divergence analysis.

time and track each other closely to then begin a decreasing phase, this means that the model is rigid at the beginning, but then it starts to converge and improve the the probability distribution assigned to the data. The free-energy gap $F(\text{data}) - F(\text{fantasy})$ oscillates around zero without fully converging.

Figure 2 examines the CD update in detail. The standard deviation of the positive-phase correlation $\langle \mathbf{xz} \rangle_{\text{data}}$ and the negative-phase correlation $\langle \mathbf{xz} \rangle_{\text{model}}$ converge from epoch 10 onward. The relative gap $|\text{data} - \text{model}|/\text{data}$ is largest in the first 10-20 epochs and then drops near zero, indicating that the fantasy particles become representative of the learned distribution early in training.

Figure 3 shows the evolution of the 12 weight filters $\mathbf{W}_{:,j}$ across epochs 1, 25, 50, 100, and 150. At epoch 1 the filters are unstructured. By epoch 25 elongated features begin to emerge, and the patterns stabilize by epoch 100-150 into oriented edges and curved strokes characteristic of digits 0, 1, and 2. Each filter captures a distinct aspect of the input. The visible bias also converges to an image reflecting the average pixel intensity across the three digit classes.

The generative quality is tested by running a Gibbs chain initialized from real samples of each digit class. Figure 4 shows that the predicted class remains constant across all 100 Gibbs steps for all three starting digits. The chain does not transition between classes, demonstrating that the model has learned stable, well-separated energy minima.

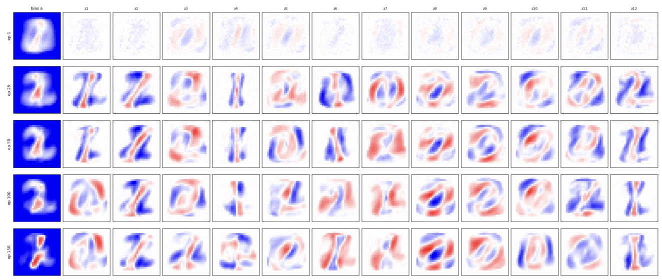

FIG. 3. Evolution of weight filters and bias during training ($L = 12$). Each row corresponds to a different epoch (1, 25, 50, 100, 150). The leftmost column shows the visible bias. Columns 2-13 show the 12 hidden unit filters. Structure emerges progressively and stabilizes by epoch 100.

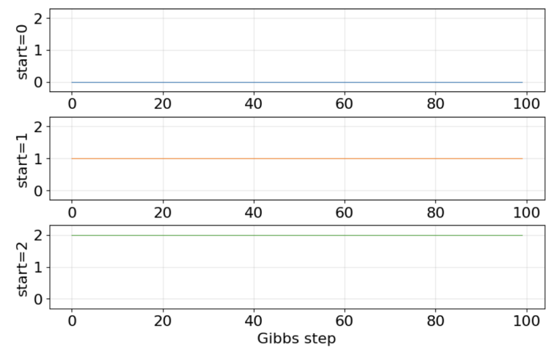

FIG. 4. Digit classification along a Gibbs chain. Three chains are initialized from real samples of digits 0, 1, and 2. The predicted class remains constant across 100 Gibbs steps, demonstrating stable, well-separated modes in the learned distribution.

Impact of Visible Bias Initialization on RBM Convergence

This section analyses how the visible bias initialization strategy proposed by Hinton (2010) affects the training of Restricted Boltzmann Machines (RBMs) when Ising Spins are not involved. Usually, visible biases a_i are set to zero before training, meaning the network must learn both the basic shape of the data and the complex connections at the same time. On the other hand, Hinton's method sets each bias using the training data statistics using equation 6. This formula puts the basic frequency of the data directly into the visible layer before the actual training starts. This effect is visually clear at epoch zero, as shown in Figure 5: while the standard zero-initialization is just an empty, uniform grid, Hinton's initialization already captures the average shape and silhouette of the input digits.

To test this idea, we trained two RBM models under identical conditions using a subset of the MNIST dataset (digits 0, 1, and 2). We tracked the training process for 50 epochs using five key metrics: reconstruction error, pseudo-likelihood, free-energy gap, and log-likelihood es-

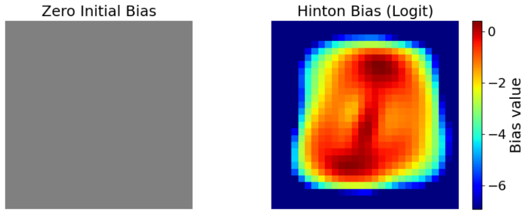

FIG. 5. A 28×28 pixel visualization at epoch zero. It shows that zero-initialization is just an empty grid, while Hinton's initialization already shows the average shape of the digits.

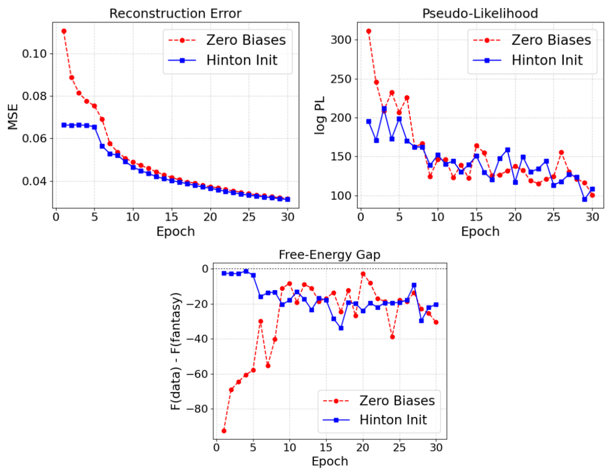

FIG. 6. Learning Curves: a plot showing the metrics over the 50 epochs, highlighting the constant performance gap between the two models

timated via Annealed Importance Sampling (AIS).

The results show that Hinton's initialization gives the model a clear and immediate advantage. As plotted on Figure 6, right from epoch zero, the reconstruction error is much lower, and the AIS log-likelihood is significantly higher than the standard zero-initialization. This initial advantage lasts through the whole training process, leading to faster convergence and better numerical stability. The free-energy gap also confirms that this method does not cause overfitting. Instead, it allows the hidden layers to immediately focus on learning complex patterns rather than wasting time learning the basic layout of the input images.

Energy Barriers in CD Sampling

To study why the Contrastive Divergence (CD) algorithm rarely switches between different digit representations, we implemented a modified forced transition between two classes (e.g., digit "1" and digit "2"). The visible layer was gradually interpolated at each step τ as

$$v_\tau = (1 - \alpha_\tau)\,\text{prototype}_1 + \alpha_\tau\,\text{prototype}_2,$$

with $\alpha_\tau \in [0, 1]$. For each intermediate visible configuration, the hidden units were sampled from $P(h|v_\tau)$ and the joint RBM energy (Eq. 1) was computed. Energy

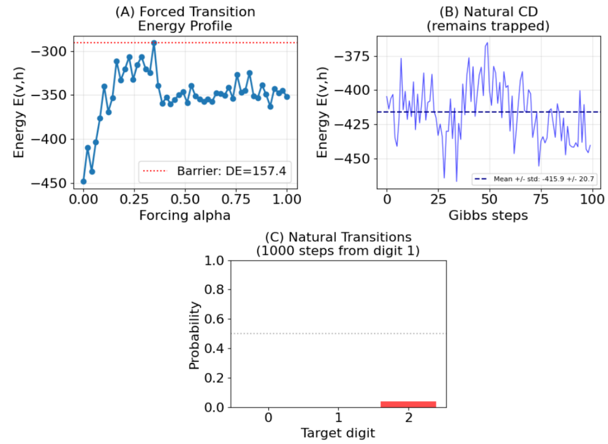

FIG. 7. Energy profiling and quasi-ergodicity analysis. (A) Joint energy $E(v, h)$ during the forced transition from digit 1 to digit 2. (B) Joint energy during unconstrained, natural CD sampling. (C) Empirical transition probabilities after 1000 Gibbs steps starting from digit 1.

barriers separate digit classes (Fig. 7 A): intermediate configurations occupy high-energy regions inaccessible to standard CD sampling (Fig. 7 B). Consequently, the Markov chain remains trapped near its initialization, explaining quasi-ergodicity. Our forced-transition method successfully crosses these barriers (Fig. 7 C), a transition a natural CD chain cannot spontaneously achieve.

Effect of Hyperparameter variation

The number of hidden units L is varied from 1 to 24. Figure 8 shows that reconstruction error decreases from about 0.060 to 0.025 as L increases. The gradient RMS decreases for all L, although larger models converge more slowly. The free-energy gap fluctuates around zero, except for $L = 4$, which exhibits unstable oscillations. Overall, larger L improves reconstruction quality.

The number of Gibbs steps N_t used in Contrastive Divergence is varied across $N_t \in \{1, 2, 5, 10, 20\}$. Figure 9 shows that $N_t = 1$ gives worse reconstruction error (~ 0.037), while all models with $N_t \geq 2$ converge near 0.032-0.033. This suggests that a single Gibbs step produces a biased estimate, whereas two steps could be already sufficient. Gradient RMS behaves similarly for all values of N_t, although $N_t = 20$ exhibits wide oscillations at the end.

The effect of L1 regularization is studied by varying the penalty coefficient $\gamma \in \{0, 0.0001, 0.001, 0.01, 0.1\}$. Figure 10 shows that values up to $\gamma = 0.01$ converge near 0.030, while $\gamma = 0.1$ fails to learn and remains flat near 0.068 throughout training. This is confirmed by RMS.

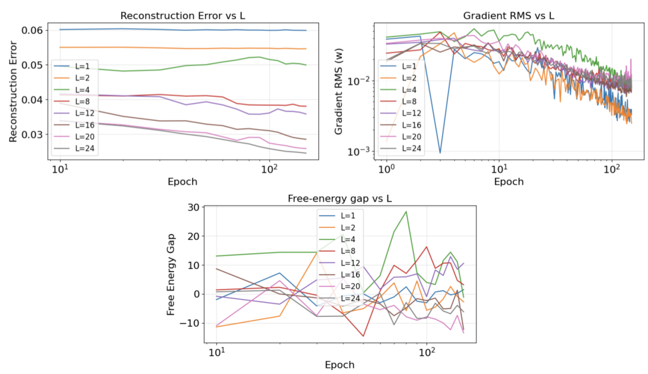

FIG. 8. Metrics for the analysis on the variation of L parameter

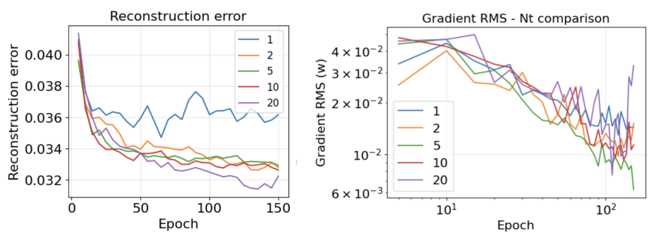

FIG. 9. CD steps variation analysis

All models with $\gamma \leq 0.01$ show the expected decay toward 10^{-2}, while $\gamma = 0.1$ maintains a constant gradient RMS near 10^{-1} with no sign of convergence.

The optimizers SGD and RMSprop, are compared with learning rates $\mathrm{lr} \in \{0.001, 0.005, 0.010\}$. Figure 11 shows that both optimizers converge to similar final reconstruction errors around 0.034-0.035. SGD with small learning rates converges more slowly, while RMSprop_lr005 converges faster and performs overall slightly better. RMS trajectories reach comparable final values near 10^{-2}, but RMSprop_lr010 and lr005 exhibits higher gradient values in the latter training phase.

Figure 12 shows the effect of SPINS encoding. With SPINS=False, reconstruction error remains at approximately 0.03 and free-energy gap stays near zero, confirming stable learning. In stark contrast, SPINS=True produces reconstruction error of 0.25, indicating failure to learn. This likely stems from incompatibility between

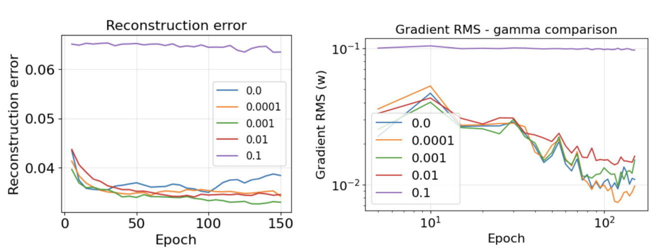

FIG. 10. Effect of L1 regularization with different values of γ

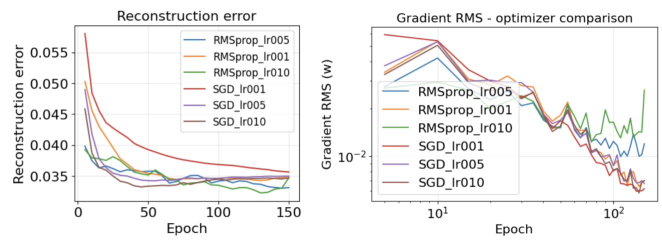

FIG. 11. Analysis of the variation of the optimizer at different learning rates

FIG. 12. Effect of SPINS encoding

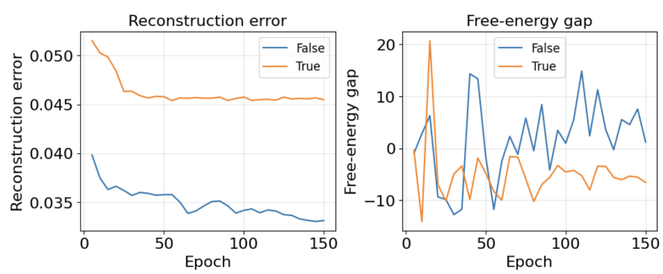

FIG. 13. Effect of POTTS encoding

tanh activations and our continuous gradient strategy (Eq. 5), disrupting variance reduction.

Figure 13 compares POTTS=False with POTTS=True. The reconstruction error converges to approximately 0.033 vs 0.045 respectively. POTTS underperformance reflects insufficient data to justify categorical units' complexity—the binary framework provides better bias-variance trade-off with 300 samples per class. While the model learns in both cases, POTTS=False offers superior reconstruction quality

Energy barriers between digit classes

The trained model develops a distinct energy landscape with two deep minima and a shallow one. Figure 14 shows the mean free energy for each class: classes 0 and 2 occupy deep minima at approximately +120, while class 1 has a shallower minimum near zero. This indicates that classes 0 and 2 are energetically more stable. Figure 15 displays the energy barriers ΔF between all pairs of classes. Transitions from class 1 to classes 0

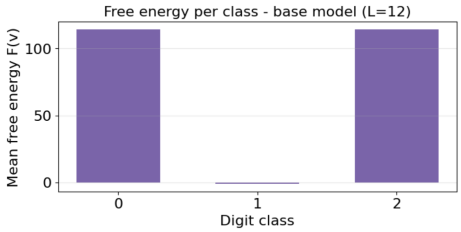

FIG. 14. Mean free energy for each class

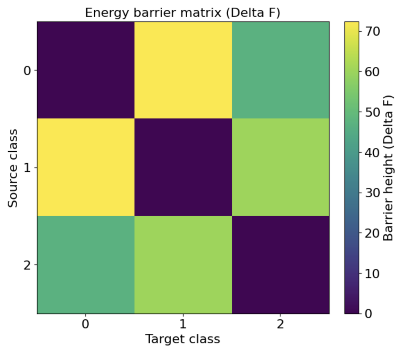

FIG. 15. Matrix of the energy barrier between digits

and 2 require to cross hights of about 70 and 40-45 units respectively. Conversely, the barrier between classes 0 and 2 is lower (30-35 units). This reveals that the model learns to treat classes 0 and 2 as similar, while class 1 remains energetically separated.

CONCLUSIONS

In this work, we implemented and evaluated a Restricted Boltzmann Machine trained on a subset of the MNIST dataset (digits 0, 1, 2), following Hinton's practical guidelines [1]. Our results show that the visible bias initialization proposed by Hinton in Eq.6 improves RBM training with respect to standard zero-initialization. The analysis of energy barriers also explains the behaviour of Contrastive Divergence sampling. Different digit classes are separated by high-energy regions, making spontaneous transitions very unlikely. As a result, the Markov chain remains trapped inside local minima and mainly explores configurations belonging to the same digit class. Among the structural extensions explored, the binary framework (Spins = False, Potts = False) consistently outperforms both the Ising-spin and Potts-unit variants in terms of reconstruction error and training stability.

The hyperparameter search identified the optimal configuration: $L = 16$, $N_t = 10$, $\gamma = 0.0001$, RMSprop with lr = 0.001, SPINS=False and POTTS=False. Using this configuration, the model was trained with 5-fold cross-validation to ensure robust generalization; the resulting

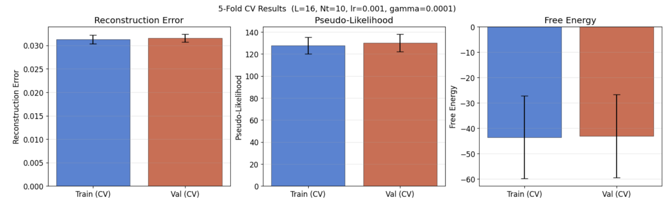

FIG. 16. Metrics of the training with cross validation.

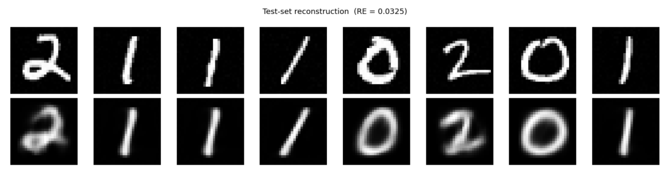

FIG. 17. Reconstruction performance on test with the cross validation model

training and validation metrics are shown in Fig. 16, and the reconstruction quality on the test set is illustrated in Fig. 17.

All code, data pipelines, and analysis notebooks produced for this project are publicly available at the following GitHub repository: `https://github.com/francesco-erario/RBM_analysis_on_MNIST.git`.

AUTHORS CONTRIBUTIONS

Francesco Erario and Laura Mazzilli implemented the model in the file rbm.py. Regarding the sections of the assignment: Francesco Erario was responsible for the section on Hyperparameter Variation, Andrea Lorenzetto for the section on Energy Barriers in CD Sampling, Laura Mazzilli for the section on Visible Bias Initialization, and Nicola Zuccotti for the Training Analysis section. The initial theoretical analysis and was carried out by all authors.

[1] G. E. Hinton, A Practical Guide to Training Restricted Boltzmann Machines. (2010)